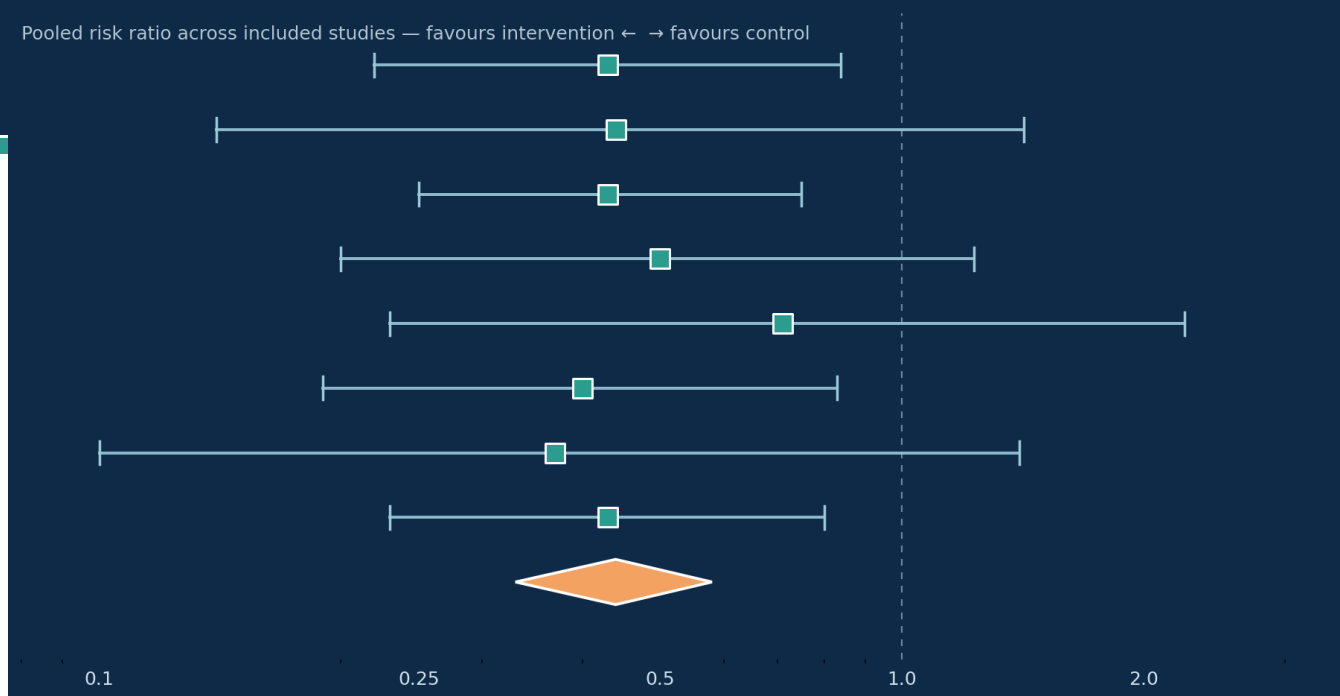


Clinical Research Assistant

An AI co-investigator for systematic reviews, meta-analyses, and evidence synthesis — built for hospital research offices, academic medical centres, and clinical research organisations.



Sample forest plot output — every effect estimate traces back to a real database hit.

A high-quality systematic review takes a small team **6–12 months**: scoping the question, building search strategies, screening thousands of abstracts, extracting data, running the statistics, assessing bias, and writing it up. Most of that work is repetitive, methodology-bound, and easy to get wrong in ways that don't show up until a reviewer rejects the manuscript.

The **Clinical Research Assistant** compresses the methodological scaffolding into a guided, auditable workflow — without ever fabricating evidence. Your investigators stay in the driver's seat for clinical judgment. The assistant handles the parts that should never have been manual in the first place.

Five guided workflows plus a continuous-monitoring service. Each one launches from natural language or a slash command and produces a structured, citation-anchored output.

1 Meta-analysis workflow

End-to-end from a research question to a forest plot. The assistant walks the user through **PICO** confirmation, runs the literature search across configured databases, presents candidates for inclusion/exclusion, captures the extraction table, then computes the pooled effect estimate with heterogeneity statistics and renders a publication-grade forest plot inside an isolated compute sandbox.

Outcome — *what used to be a multi-week analyst engagement becomes a single guided session. Every PMID traces back to a real database hit.*

2 Search-strategy builder

Constructs Boolean queries with MeSH terms, field tags, and database-specific syntax (PubMed today; Embase / Cochrane on the roadmap). Iterates on **broaden / tighten** suggestions with the user until the strategy is registration-ready.

Outcome — *defensible, reproducible search strings in minutes instead of half a day, with full iteration history preserved.*

3 Systematic-review protocol drafter

Generates a **PRISMA-P-aligned** protocol skeleton — background, objectives, eligibility, search methods, screening plan, data items, risk-of-bias plan, synthesis approach — ready to deposit in **PROSPERO**.

Outcome — *a protocol-ready first draft your methodologist edits, not writes from scratch.*

4 Risk-of-Bias assessor

Supports the major instruments: **RoB 2** (RCTs), **ROBINS-I** (non-randomised), **Newcastle-Ottawa** (observational), **QUADAS-2** (diagnostic accuracy). Walks domain-by-domain, asks the judgement questions, and produces the structured assessment table.

***Outcome** — consistency across reviewers and a clean audit trail of why each judgement landed where it did.*

5 General clinical Q&A

Background reading, definitions, guideline navigation. Uses web and Wikipedia tools, but is **hard-prevented from quoting effect sizes, PMIDs, or guideline citations from training data** — a regex-level validator blocks unsupported clinical claims before they reach the user.

***Outcome** — safe-by-default exploratory Q&A that won't seed a manuscript with hallucinated evidence.*

6 Living-review watches (scheduled monitoring)

Pin a PICO and a search strategy as a **watch**, set a cadence (daily / weekly / monthly), and the assistant re-runs the search on schedule, diffs against the baseline corpus, triages newly published papers against the original PICO, and **notifies the team when something materially shifts the evidence base**.

***Outcome** — living reviews stop drifting out of date. Guideline committees and HTA bodies get alerted to practice-changing evidence as it lands.*

What makes it different

Differentiator	Why it matters
Anti-hallucination enforced in code	A regex validator and tool-gated PMID rule block fabricated citations even if the underlying model drifts. Your investigators can trust the bibliography.
Source-agnostic search across paid + open databases	PubMed, Europe PMC, Embase, Cochrane Library, Scopus, Web of Science — all live behind a pluggable PaperSource interface and fanned out in parallel with cross-source de-duplication. One admin panel for credentials and rate limits.

Local research cache + RAG over pulled content	Every abstract, full-text article, and extraction table you pull lands in the local research store. A retrieval-augmented pipeline searches that store first, so the same paper isn't re-fetched — or re-billed — across reviews. Pull paid content once; reuse it across the entire research programme.
Sandboxed statistical compute	All Python analysis runs inside a Docker container with networking disabled, read-only inputs, and capped CPU/memory. No host access; no leakage.
Workflow-gated tools	The assistant can't skip ahead — no meta-analysis before a confirmed PICO; no full text before studies are selected. The methodology is the guardrail.
Flexible, role-aware deployment	Deploy on-premises or in your cloud — the FastAPI core runs anywhere from a single workstation to a managed container platform. Role-based access control gates each workflow per user, and onboarding a new researcher is a one-step provisioning action that scopes their search-API keys, model access, and visibility into shared review threads from day one.
Full conversation persistence	Every turn — PICO, included studies, extraction table, generated code, plot — is stored. Reproducing an analysis a year later is just re-opening the thread.

What it costs to run

Two variable costs to plan around: LLM tokens on AWS Bedrock and web-search calls on Tavily. Everything else — paper database access, paper storage, the FastAPI app, the sandbox — sits on infrastructure you already own. Paid bibliographic databases (Embase, Scopus, Cochrane, etc.) are assumed to be covered by your existing institutional subscriptions.

ACTIVE RESEARCH DAY (PER USER)

\$200 – \$300

Envelope for a productive day of meta-analysis work: PICO refinement, multi-database search, screening + extraction across dozens of papers, statistical analysis with code generation, and forest-plot rendering. Covers **Bedrock model tokens + Tavily web search**.

INCREMENTAL COST OF A RE-OPEN

≈ \$0

Re-opening an existing thread to inspect, audit, or re-export the analysis costs only the few tokens of the current turn. Past tool results (papers, extractions, generated code, plots) are already persisted in the local store.

How the envelope is built

An active meta-analysis session is token-heavy because the model reads abstracts, reasons over extraction tables, writes statistical code, and reviews its own outputs. On Claude Sonnet-class pricing, a serious day of work lands in the **\$200–\$300** range — comfortably below the cost of an equivalent human-analyst day. Light browsing, background reading, or quick definitional Q&A sessions run at a small fraction of that figure.

Tavily web search is consumed only by the General Q&A and protocol-drafting flows; the clinical search itself runs against PubMed and Europe PMC and does **not** consume Tavily credits.

Why the local store keeps costs flat

Pulled study material — abstracts, MeSH lookups, fetched PMC full text, generated extraction tables — is persisted in CRA's local store. The next time the same study or the same PICO is touched (e.g. by a living-review watch, by a sibling project, or by a re-run on updated data), the assistant reads from the local cache instead of re-issuing API calls. **The marginal cost of revisiting prior work approaches zero.**

This matters most for teams running **multiple parallel reviews** in overlapping areas — cardiovascular, oncology, infectious disease — where the same landmark trials show up across many questions. The first review pays for the lookup; every subsequent review reuses it for free.

Cost-control lever	How it keeps spend predictable
Local study cache	Re-use of pulled abstracts / full text across reviews — no repeat API spend.

Workflow-gated tools	Expensive tools (sandbox, full-text fetch) are unreachable until the workflow stage needs them — no idle calls.
Per-turn ceilings	max_model_requests and agent_timeout_seconds bound runaway loops to a known maximum cost.
Configurable model tier	Drop to a cheaper Sonnet / Haiku variant per workflow when full Opus reasoning isn't required.

On the roadmap

Beyond the v1 capability set, the architecture was built to absorb these without rebuilding the core. Each is a tractable next increment, not a moonshot.

Automated clinical data collection

eCRF designer — generate a draft electronic case-report form directly from a study protocol, with field types, validation rules, and skip logic mapped to the protocol's data items. Reduces the weeks-long round-trip between investigator and data manager at study start-up.

Source-document extraction — point the assistant at structured exports from your EHR / registry / trial-management system and have it populate the extraction table for retrospective studies or patient-level meta-analyses, with an audit trail of which source row produced which output cell.

Patient-facing research handouts

Generate **plain-language summaries** of a study's objectives, what participation involves, and what the early evidence suggests — at a configurable reading level, in the patient's preferred language. Designed for **research participant recruitment**, **shared decision-making** conversations, and **return-of-results** obligations under modern IRB / ethics guidance. The lay summary and the manuscript share a single evidence base, so they cannot drift apart.

Research-gap analysis specialist

Given a body of literature, surface where the evidence base is thin — by population, intervention, outcome, geography, or study design — so investigators can prioritise the next grant proposal where the field actually needs new data.

Group-level living-review subscriptions

Group watches with quorum-based notification rules — designed for guideline committees and HTA bodies who need consensus signalling on practice-changing evidence rather than per-user alerts.

If your researchers spend more time finding and formatting evidence than interpreting it, this is what you point them at. They keep the clinical judgement. The assistant absorbs the methodological choreography — and writes it down as it goes.